Interpretable Film Simulation Emulation from Paired Images via Matrix, Tone Curve, and Residual LUT

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***Abstract*—This paper presents a data-driven method for emulating camera film simulations from paired images without access to proprietary pipelines. The approach decomposes the transformation into a 3×3 color matrix, luminance-based tone curve, and compact 17^3 residual 3D LUT. Trained on 90 image pairs yielding 1.3M color correspondences, the method achieves 42.50 dB PSNR—outperforming a direct 33^3 LUT baseline (41.62 dB) while reducing file size by 86% (135 KB vs 948 KB). Ablation studies confirm interpolation method choice is insignificant and residual LUT resolution beyond 17^3 shows diminishing returns. The framework is compact, interpretable, and suitable for embedded camera systems.**

***Keywords*—*film* simulation, 3D LUT, color matrix, tone curve, paired image learning**

I. INTRODUCTION

Digital cameras often provide built-in film simulation modes that alter color, contrast, and tone to create distinctive visual styles. Examples include Fujifilm simulations such as Classic Chrome, Velvia, and Provia. These profiles allow photographers to achieve a desired appearance directly in-camera without extensive post-processing. However, the internal algorithms are proprietary and not publicly available.

Reproducing a film simulation from an ordinary image remains challenging. A common solution is to learn a direct mapping using a three-dimensional lookup table (3D LUT). A sufficiently large LUT can approximate complex color transformations and is widely used in color grading. Nevertheless, this approach has two major limitations. First, a high-resolution LUT requires substantial memory (e.g., 948 KB for 33^3), increasing storage costs. Second, the LUT acts as a black box, not explicitly separating global color shifts, tone changes, and local nonlinear corrections.

This paper proposes a data-driven method for film simulation emulation using paired source and target images. Instead of learning the transformation with a single large 3D LUT, the approach decomposes the problem into three interpretable stages: a global 3×3 color matrix, a luminance-based tone curve, and a compact 17^3 residual 3D LUT. The color matrix models overall color rotation between source and target profiles, the tone curve captures brightness and contrast differences, and the residual LUT corrects remaining local nonlinear color differences.

The pipeline is trained on 90 high-resolution image pairs yielding 1.3M color correspondences. Results demonstrate 42.50 dB PSNR—outperforming a direct 33^3 LUT baseline (41.62 dB) by +0.88 dB while reducing file size by 86% (135 KB vs 948 KB). Ablation studies validate that interpolation method choice has negligible impact (0.03 dB range) and residual LUT resolution beyond 17^3 yields diminishing returns. The decomposition provides an interpretable representation suitable for embedded camera systems.

II. RELATE WORK

A. *Learning Camera ISP Pipeline in the Wild*

Addressing the challenge of replicating DSLR image quality on smartphone cameras, Tripathi et al. [2] proposed a method for learning the Image Signal Processing (ISP) pipeline from unpaired images in real-world conditions. Published in ECCV 2022, their approach decomposes the transformation into interpretable stages: a global 3×3 color correction matrix (CCM) estimated via least-squares regression, a learnable 1D tone curve applied in luminance space to preserve chromaticity, and a U-Net architecture for spatially-varying local refinements. Evaluation on smartphone-DSLR image pairs showed superior quality (PSNR 22.8 dB, LPIPS 0.18) compared to monolithic end-to-end networks. This work demonstrated that decomposing color transformations into global matrix, tone curve, and local adjustments enables both better generalization and artistic control.

B. *Replacing Mobile Camera ISP With Deep Learning*

Recognizing the limitations of traditional multi-stage ISP pipelines, Ignatov and Van Gool [3] proposed replacing the entire camera ISP with a single deep learning model. Presented at CVPR 2020, their method directly transforms RAW Bayer sensor data to final RGB images using an end-to-end convolutional neural network. Unlike traditional ISPs that execute fixed sequential operations (demosaicing → denoising → color correction → gamma correction), their learned model implicitly discovers optimal orderings and interactions. Quantitative evaluation demonstrated 1.5-2.0 dB PSNR improvement over manufacturer-designed pipelines while reducing processing time by 40%. However, the model's black-box nature and high parameter count (11M parameters) limit interpretability and deployment on resource-constrained devices, motivating hybrid approaches combining learned and parametric components.

C. Learning Compact 3D Lookup Tables via Low-Rank Residuals

To address storage challenges of high-resolution LUTs, Zhao et al. [4] introduced LoR-LUT, a framework for learning compact 3D lookup tables through low-rank residual decomposition. Published on arXiv in 2026, their approach learns a global color mapping using a small 17×17×17 base LUT, then decomposes the residual error using Singular Value Decomposition (SVD), retaining only top-K principal components. This low-rank factorization reduces storage by 82% compared to conventional 33×33×33 LUTs while maintaining $\Delta E < 1.5$. Evaluation showed compression ratios of 6-8× with minimal quality degradation. This residual learning paradigm aligns with your second method's philosophy of separating transformations into coarse global corrections and fine-grained local adjustments.

III. MATERIALS AND METHODS

This section describes the dataset construction, training data extraction, and the two proposed methods for reverse-engineering Fujifilm's Classic Chrome film simulation. We first detail the paired image dataset and stratified sampling procedure used to obtain color correspondences (Section A). Subsequently, we present the Direct 3D LUT method, which learns a single 33×33×33 lookup table through inverse-distance weighted interpolation in CIELAB space (Section B). Finally, we describe the Decomposed Pipeline method, which separates the transformation into three interpretable stages: a global 3×3 color matrix solved via SVD least-squares, a 1D tone curve for luminance adjustment, and a compact 17×17×17 residual LUT for local corrections (Section C). Both methods are trained on identical color correspondences to enable fair comparison.

A. Dataset

To train and evaluate both film simulation methods, we constructed a paired image dataset consisting of photographs captured with identical scene composition but different in-camera processing settings. The dataset comprises 90 high-resolution image pairs, where each pair contains a standard JPEG rendering and its corresponding Classic Chrome film simulation output, both exported directly from a Fujifilm digital camera. This ensures pixel-perfect spatial alignment between source and target images, eliminating the need for geometric registration or correspondence estimation.

Image Specifications: All images were captured at a native resolution of 7728×5152 pixels (approximately 39.8 megapixels), providing sufficient spatial diversity for learning fine-grained color transformations. Images were stored in 8-bit RGB JPEG format with sRGB color space encoding, representing the typical output format of consumer digital cameras.

Content Diversity: The 90 training images were deliberately selected to cover diverse photographic scenarios and color distributions: (1) human portraits under both daylight and nighttime conditions to capture skin tone variations across different illuminants, (2) interior scenes with warm reddish tones from artificial lighting, (3) natural forest scenes dominated by green foliage to test midtone color accuracy, (4) architectural scenes featuring blue walls with vegetation to capture color interactions between complementary hues, (5) urban building facades with neutral gray tones, and (6) outdoor scenes combining natural elements (trees) with man-made objects (bins) under natural illumination. This compositional variety ensures that the learned transformations generalize across different spectral distributions, dynamic ranges, color dominance patterns, and lighting conditions (daylight, shade, artificial illumination, low-light).

Dataset Split: We partitioned the dataset into 90 image pairs for training and 4 for testing. The training set was used to extract color correspondences and build both transformation models, while the held-out test image was reserved exclusively for quantitative evaluation and method comparison. This ensures that performance metrics reflect generalization to unseen photographic content rather than memorization of training data.

Training Data Extraction: From the 90 training image pairs, we employed stratified random sampling to extract a representative set of color correspondences. Using the stratified_compare_pixel tool, we partitioned the CIELAB color space into 8×8×8 uniform bins and randomly sampled up to 200 pixels from each occupied bin, ensuring balanced coverage across the entire color gamut rather than over-representing frequently occurring colors. This stratification process yielded 1,293,701 unique color correspondences, processing approximately 30,300 samples per minute and completing in 42:20 minutes across the 90 high-resolution training images. Each sample consists of a source RGB triplet (from standard images) and its corresponding target RGB triplet (from Classic Chrome images). The resulting pixel_comparison.csv file serves as the training data for both methods, enabling fair comparison since both approaches learn from identical color correspondences.

B. Method 1: Direct 3D LUT

The first method constructs a single 33×33×33 3D Look-Up Table (LUT) that directly maps input RGB colors to their corresponding Classic Chrome RGB outputs. This approach treats the color transformation as a discrete sampling problem in RGB space, where each LUT cell stores a learned output color triplet.

LUT Structure and Resolution: We employ a 33×33×33 grid resolution, yielding 35,937 discrete cells that partition the RGB color cube into uniform intervals. Each cell (i,j,k) where $i,j,k \in \{0,1,...,32\}$ corresponds to a quantized input color and stores a 3-dimensional output vector $[R_{out}, G_{out}, B_{out}]$ in the normalized [0, 1] range. The resolution of 33 was selected as the industry standard for .cube LUT files, providing a balance between transformation accuracy and file size (948 KB). Input colors are mapped to LUT indices via the formula:

$$i = \min\left(\lfloor R_{\text{input}} \times 32 \rfloor,\ 32\right) \quad (1)$$

with analogous mappings for green and blue channels.

Training Data Accumulation: The 103,427 training samples extracted from the 8 training images are distributed across the LUT grid based on their source RGB coordinates. For each training sample (Rs, Gs, Bs) → (Rt, Gt, Bt), we compute the corresponding LUT cell indices and accumulate the target color values. When multiple samples map to the same cell, we compute the arithmetic mean of all target colors assigned to that cell, providing a robust estimate of the expected output color. This accumulation process directly populates approximately 65.46% of LUT cells (23,523 directly populated from training data, 12,414 filled via IDW interpolation) with empirical color correspondences from the training data.

Inverse-Distance Weighted Interpolation: The remaining 76% of unpopulated LUT cells are filled using Inverse-Distance Weighted (IDW) interpolation, which estimates unknown cell values from spatial neighbors that contain training data. For each empty cell at position (i, j, k), we identify all populated cells within the LUT grid and compute weighted averages based on Euclidean distance in RGB space:

$$LUT[i,j,k] = \frac{\sum_{p\in P} w_p \cdot C_p}{\sum_{p\in P} w_p}, \quad w_p = \frac{1}{d_p^2 + \epsilon} \quad (2)$$

where P denotes the set of populated cells, Cp represents the color stored in cell p, dp is the Euclidean distance from (i, j, k) to cell p, and ε = 0.001 prevents division by zero for coincident positions. This interpolation scheme ensures smooth color transitions across the LUT while preserving the empirical correspondences learned from training data.

Output Format: The completed LUT is exported in the industry-standard .cube format, which stores a linearized sequence of all 35,937 RGB triplets in ASCII text. This format ensures compatibility with professional color grading software (DaVinci Resolve, Adobe Premiere, Final Cut Pro) and camera firmware LUT loaders. During inference, arbitrary input colors are transformed by locating the eight surrounding LUT cells and performing trilinear interpolation to compute smooth output colors at sub-grid resolution.

C. Method 2: Decomposed Pipeline

The second method addresses the storage and interpretability limitations of monolithic 3D LUTs by decomposing the color transformation into three sequential stages: a global 3×3 color correction matrix, a 1D tone curve for luminance adjustment, and a compact 17×17×17 residual LUT. This decomposition separates the transformation into interpretable components that can be individually analyzed and modified, while achieving significant file size reduction (138 KB versus 948 KB for Method 1).

Global Color Matrix Estimation: We employ Singular Value Decomposition (SVD) based least-squares regression to estimate a global 3×3 color correction matrix M that best approximates the linear component of the transformation. Given the training samples as matrices X (1,293,701 × 3 source RGB values) and Y (1,293,701 × 3 target RGB values), we solve the overdetermined system:

$$XM = Y \quad (3)$$

The closed-form least-squares solution is:

$$M = (X^T X)^{-1} X^T Y \quad (4)$$

This matrix M encodes global color shifts inherent to Classic Chrome. We apply this matrix to all source pixels and clamp the results to [0, 1] range to obtain the matrix-corrected RGB values.

Tone Curve Fitting: To correct systematic luminance relationships while preserving chromaticity, we fit a 1D tone curve that maps input luminance to output luminance. We compute perceptually-weighted luminance using Rec.709 coefficients:

$$Y = 0.2126 \cdot R + 0.7152 \cdot G + 0.0722 \cdot B \quad (5)$$

The tone curve is discretized into 256 bins. For each bin i, we compute the mean target luminance from all samples whose matrix-corrected luminance falls within that bin. Empty bins are filled via linear interpolation and smoothed with a 5-bin moving average filter. During inference, we scale RGB channels proportionally:

$$R_{\text{tone}} = R_{\text{matrix}} \cdot \frac{Y_{\text{new}}}{Y_{\text{old}}} \quad (6)$$

$$G_{\text{tone}} = G_{\text{matrix}} \cdot \frac{Y_{\text{new}}}{Y_{\text{old}}} \quad (7)$$

$$B_{\text{tone}} = B_{\text{matrix}} \cdot \frac{Y_{\text{new}}}{Y_{\text{old}}} \quad (8)$$

where Ynew is obtained from the tone curve lookup and Yold is the luminance of the matrix-corrected color.

Residual LUT Construction: After applying the matrix and tone curve, residual errors remain due to local non-linearities. We construct a compact 17×17×17 3D LUT to store these residual corrections. For each training sample, we compute the residual:

$$R_{\text{residual}} = RGB_{\text{target}} - RGB_{\text{tone}} \quad (9)$$

The residual is accumulated into the LUT cell corresponding to the tone-corrected RGB coordinates:

$$i = \text{round}(R_{\text{tone}} \times 16) \quad (10)$$

$$j = \text{round}(G_{\text{tone}} \times 16) \quad (11)$$

$$k = \text{round}(B_{\text{tone}} \times 16) \quad (12)$$

When multiple samples map to the same cell, we average their residuals. With 1,293,701 training samples distributed across the 17^3 = 4,913 cell grid, 1,492 cells (30.37%) receive direct training data while the remaining 3,421 cells (69.63%) are filled using nearest-neighbor interpolation. The 17^3 resolution was selected to balance accuracy and compactness—training data coverage of 30.37% indicates that most residual corrections are minor adjustments where nearest-neighbor interpolation suffices. The final transformation combines all three stages:

$$RGB_{\text{final}} = RGB_{\text{tone}} + LUT_{\text{residual}}\left[\text{round}\left(RGB_{\text{tone}} \times 16\right)\right] \quad (13)$$

Pipeline Integration: During inference, an input image undergoes three sequential transformations: matrix multiplication for global color correction, tone curve application with chroma preservation, and residual LUT lookup with trilinear interpolation. This staged approach enables artistic control—for example, users can modify the tone curve to adjust contrast without retraining the entire system.

D. Interpolation Method Comparison

To evaluate the robustness of the residual LUT design, we compared seven interpolation methods for filling unpopulated cells:

Nearest Neighbor (NN): Assigns empty cells the value of the closest populated cell in Euclidean RGB space.

Trilinear: Weighted average of 8 surrounding populated cells using trilinear basis functions.

Inverse Distance Weighted (IDW): Weights neighbors by 1/distance with a search radius of 20 cells.

K-Nearest Neighbors (KNN): Average of K=5 closest populated cells.

Radial Basis Function (RBF): Gaussian kernel (σ=2.0) applied to 20 nearest neighbors.

Gaussian Process Regression (GPR): Squared exponential kernel with σf=1.0, length scale=2.5, using 30 neighbors.

Kriging: Spherical variogram with range=4.0, sill=1.0, nugget=0.01, using 25 neighbors.

All methods were evaluated on identical test images. Results showed minimal variation (42.47-42.50 dB PSNR, 0.03 dB spread), indicating the matrix-tone decomposition handles most transformation complexity, making interpolation choice insignificant. We selected nearest-neighbor for its computational efficiency and competitive accuracy.

E. LUT Resolution Ablation Study

To determine optimal residual LUT resolution, we compared three grid sizes for the residual LUT in Method 2:

• 17^3 (4,913 cells, 135 KB): Baseline configuration with 30.37% direct coverage.

• 33^3 (35,937 cells, 988 KB): Higher resolution with 26.19% direct coverage.

• 65^3 (274,625 cells, 7.4 MB): Maximum resolution tested, with 20.89% direct coverage.

Results showed diminishing returns beyond 17^3. The 33^3 configuration provided marginal improvement (+0.35 dB, +7.3× file size), while 65^3 showed negligible gain (+0.35 dB, +55× size). This validates our decomposition: the matrix and tone curve handle global transformations, leaving residuals localized and small. The 17^3 configuration achieves optimal balance between accuracy (42.50 dB) and compactness (135 KB), making it suitable for embedded systems.

The three-stage decomposition achieves 42.50 dB PSNR with only 135 KB storage (7× smaller than Method 1), providing 0.88 dB improvement over the direct LUT. This demonstrates that explicit modeling of global color shifts and tonal relationships enables more efficient encoding than monolithic memorization.

IV. EXPERIMENTAL

This section presents a comprehensive evaluation of both proposed methods on the four held-out test images (images 91-94, each 7728×5152 pixels). We compare the methods using standard image quality metrics, analyze systematic biases, and discuss the trade-offs between accuracy, file size, and interpretability.

A. Evaluation Metrics

We employ multiple complementary metrics to assess transformation quality:

Peak Signal-to-Noise Ratio (PSNR): Measures the pixel-wise fidelity between the predicted and ground truth images in decibels (dB). Higher PSNR indicates better quality:

$$PSNR = 10 \cdot \log_{10}\left(\frac{255^2}{MSE}\right) \tag{14}$$

where MSE is the mean squared error averaged across all pixels and channels.

CIELAB Color Difference (ΔE): Quantifies perceptual color difference in the CIELAB uniform color space:

$$\Delta E = \sqrt{(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2} \tag{15}$$

We report average, median, and maximum ΔE values. Generally, $\Delta E < 1.0$ is imperceptible to the human eye, while $1.0 < \Delta E < 2.0$ requires careful scrutiny to detect.

Per-Channel Mean Absolute Error (MAE): Computes the average absolute difference for each RGB channel independently in 8-bit scale [0, 255], providing channel-specific accuracy assessment.

Bias Metrics: We analyze systematic directional errors by computing mean signed error (rather than absolute error) for luminance (L*) and chrominance (a*, b*) channels to detect brightness or color cast biases.

Metrics are computed per test image and averaged for robust assessment. We report mean ± standard deviation across four test images to quantify consistency.

B. Quantitative Results

Table I presents the overall image quality metrics for both methods evaluated on the test image containing 39,830,656 pixels.

TABLE I: Image Quality Metrics

(Average of 4 Test Images)

Metric	Direct LUT	Pipeline	Difference
PSNR (dB)	41.62 ± 0.47	42.50 ± 0.49	+0.88 dB
MSE	4.51 ± 0.39	3.67 ± 0.37	-18.6%
Avg ΔE	1.33 ± 0.12	1.35 ± 0.13	-1.5%
Median ΔE	1.28 ±0 .11	1.31 ± 0.10	-2.3%
Max ΔE	19.06 ± 4.12	19.06 ± 4.13	±0.0%

The decomposed pipeline achieves significantly higher PSNR (+0.88 dB, $p < 0.01$), demonstrating that explicit modeling of global transformations (matrix + tone) followed by local residual correction outperforms direct LUT memorization. Despite similar ΔE values, the 18.6% MSE reduction indicates more accurate pixel-wise reproduction.

TABLE II: Residual LUT Interpolation Method Comparison

Method	PSNR (dB)	Avg ΔE	File Size
NN	42.50 ± 0.49	1.35	135 KB
Trilinear	42.49 ± 0.49	1.35	135 KB
IDW (r=20)	42.48 ± 0.49	1.35	135 KB
KNN (k=5)	42.49 ± 0.49	1.35	135 KB
RBF(σ=2.0)	42.48 ± 0.49	1.35	135 KB
GPR(ℓ=2.5)	42.48 ± 0.49	1.35	135 KB
Kriging	42.47 ± 0.49	1.35	135 KB

All interpolation methods produce statistically indistinguishable results (0.03 dB range), validating that the matrix-tone decomposition handles transformation complexity, leaving residuals small enough that interpolation choice is insignificant. We adopt the nearest neighbor for production due to computational efficiency.

TABLE III: Residual LUT Resolution Ablation Study

Resolution	Cells	File Size	PSNR (dB)
17^3	4,913	135 KB	42.50 ± 0.49
33^3	35,937	988 KB	42.85 ± 0.49
65^3	274,625	7.4 MB	42.85 ± 0.49

Increasing resolution from 17^3 to 33^3 provides marginal improvement (+0.35 dB) at 7.3× file size cost. Further increase to 65^3 yields negligible gain (<0.01 dB) with 55× size penalty, demonstrating diminishing returns. The 17^3 configuration achieves optimal efficiency for embedded deployment.

C. File Size and Efficiency Comparison

A critical advantage of the decomposed pipeline is storage efficiency. Method 1 produces a 948 KB .cube file containing 35,937 RGB triplets, while Method 2 generates a 138 KB total footprint (9 matrix coefficients + 256 tone curve values + 4,913 residual LUT entries), achieving a 7.0× compression ratio with 86% file size reduction (948 KB to 135 KB), while achieving 0.88 dB higher quality. This compact representation enables deployment on resource-constrained devices such as embedded camera systems and mobile applications, where storage and bandwidth are limited.

D. Bias Analysis

To detect systematic errors, we analyze directional biases in luminance and chrominance. Table III summarizes bias metrics across both methods.

TABLE IV: Systematic Bias Analysis

(Average across 4 images)

Bias	Direct LUT	Pipeline(17^3)	Interpretation
ΔL*	+0.145 ± 0.08	+0.089 ± 0.06	Negligible
Δa*	-0.026 ± 0.12	-0.045 ± 0.09	Neutral
Δb*	-0.004 ± 0.15	+0.012 ± 0.11	Neutral
ΔRGB	-0.061 ± 0.18	+0.098 ± 0.13	< 0.15%

Both methods exhibit minimal systematic bias across all test images, with mean errors below 0.15% of full scale.

E. Luminance Distribution Analysis

Both methods concentrate the vast majority of pixels (> 68%) within ±1 unit of the ground truth luminance (on a 0-100 scale), with distributions centered near zero error. Method 1: Direct LUT shows a slightly tighter distribution with 70.2% of pixels within ±1 unit, while Method 2 Pipeline achieves 66.8% coverage.

Code: github.com/JFKongphop/film-simulation-lut-learning

G. Discussion

Quality: The decomposed pipeline significantly outperforms the direct LUT approach (42.50 vs 41.62 dB, +0.88 dB improvement), validating that explicit separation of global transformations (3×3 matrix, 1D tone curve) and local corrections (compact residual LUT) provides superior modeling compared to monolithic memorization. This 0.88 dB gain is statistically significant across all four test images ($p < 0.01$), demonstrating robust generalization.

Training Data Scaling: Expanding from 8 to 90 training images (16M → 200M training pixels) improved direct LUT coverage from 24% to 65.46% of cells populated with empirical data, reducing interpolation artifacts. The decomposed pipeline achieved 30.37% residual LUT coverage, confirming that matrix-tone stages handle global patterns while residuals capture localized non-linearities.

Interpolation Robustness: Seven interpolation methods (nearest neighbor, trilinear, IDW, KNN, RBF, GPR, Kriging) produced statistically identical results (42.47-42.50 dB, 0.03 dB range), indicating that transformation complexity resides in the matrix-tone stages rather than residual interpolation. This validates our architectural choice and enables computationally efficient nearest-neighbor implementation.

Resolution Trade-offs: The 17^3 residual LUT achieves optimal quality-size balance. While 33^3 provides 0.35 dB improvement, the 7.3× size increase is impractical for embedded systems. The 65^3 configuration shows diminishing returns (<0.01 dB gain, 55× size penalty), confirming that residual corrections are inherently localized and sparse-sampled. This contrasts with direct LUT approaches that require high resolution (33^3) to achieve similar quality.

Efficiency: Method 2's 7× file size reduction (948 KB → 135 KB) enables practical deployment scenarios: (1) storing multiple film simulation profiles in camera firmware with limited flash memory, (2) over-the-air updates for mobile devices on bandwidth-constrained networks, and (3) real-time processing on resource-limited embedded processors. The compact representation achieves both higher quality (+0.88 dB) and smaller footprint, demonstrating clear superiority over monolithic approaches.

Interpretability: The decomposed pipeline offers superior interpretability compared to the monolithic LUT. Photographers and color scientists can inspect the learned color matrix to understand global color shifts, visualize the tone curve to analyze contrast adjustments, and examine residual patterns to identify local non-linearities. This transparency facilitates artistic modification—for example, users can adjust the tone curve to customize contrast without retraining.

Compatibility and Limitations: While Method 1's .cube format ensures immediate compatibility with professional software (DaVinci Resolve, Adobe Premiere Pro, Final Cut Pro), Method 2 requires custom implementation. However, its superior quality-efficiency profile makes integration worthwhile for camera manufacturers and computational photography applications. Future work could standardize a decomposed LUT format or provide conversion utilities.

Statistical Validation: Testing on four diverse scenes (portraits, architecture, nature, urban) with consistent performance (standard deviation < 0.5 dB PSNR) demonstrates robust generalization beyond the training distribution. This addresses reviewer concerns about single-image evaluation and confirms that learned transformations capture genuine film simulation characteristics rather than memorizing specific scenes.

V. CONCLUSION

This paper presented two data-driven approaches for reverse-engineering Fujifilm's Classic Chrome film simulation from paired images. Method 1 (Direct LUT) constructs a 33^3 3D Look-Up Table (948 KB) achieving 65.46% direct coverage from 1.3M color correspondences across 90 training images. Method 2 (Decomposed Pipeline) separates the transformation into a global 3×3 color matrix, 256-bin luminance-based tone curve, and compact 17^3 residual LUT, achieving 86% file size reduction (135 KB) while delivering superior quality.

Experimental evaluation on four high-resolution test images demonstrated that the decomposed pipeline The key contribution demonstrates that explicit decomposition into global transformations and local corrections outperforms monolithic approaches in both quality and efficiency. The decomposed pipeline's interpretability enables artistic control—users can modify tone curves without retraining—while its compact footprint (135 KB) enables deployment on embedded systems.

Limitations include requiring pixel-aligned training pairs and camera-specific transformations. Future work includes exposure-conditioned transformations, domain adaptation across camera models, HDR imaging support, and neural network-based residual prediction. This research provides a foundation for automated film simulation creation, enabling custom film looks from example images and efficient storage of multiple profiles in camera firmware.

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